Advancing the Quest for Dark Matter (DM): A Novel Approach of Detecting Metastable Charged Particles at the Large Hadron Collider (LHC)

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Rapid Track Trigger & Reconstruction, Dark Matter Candidate

- Triggering on charged particles is currently based on energy deposition in outermost detectors
- However, there might be "disappearing tracks" of charged particles that decay upstream ($\sim 30\ cm$)
- Silicon-based track trigger enables rapid reconstruction at small radius in detector sensors

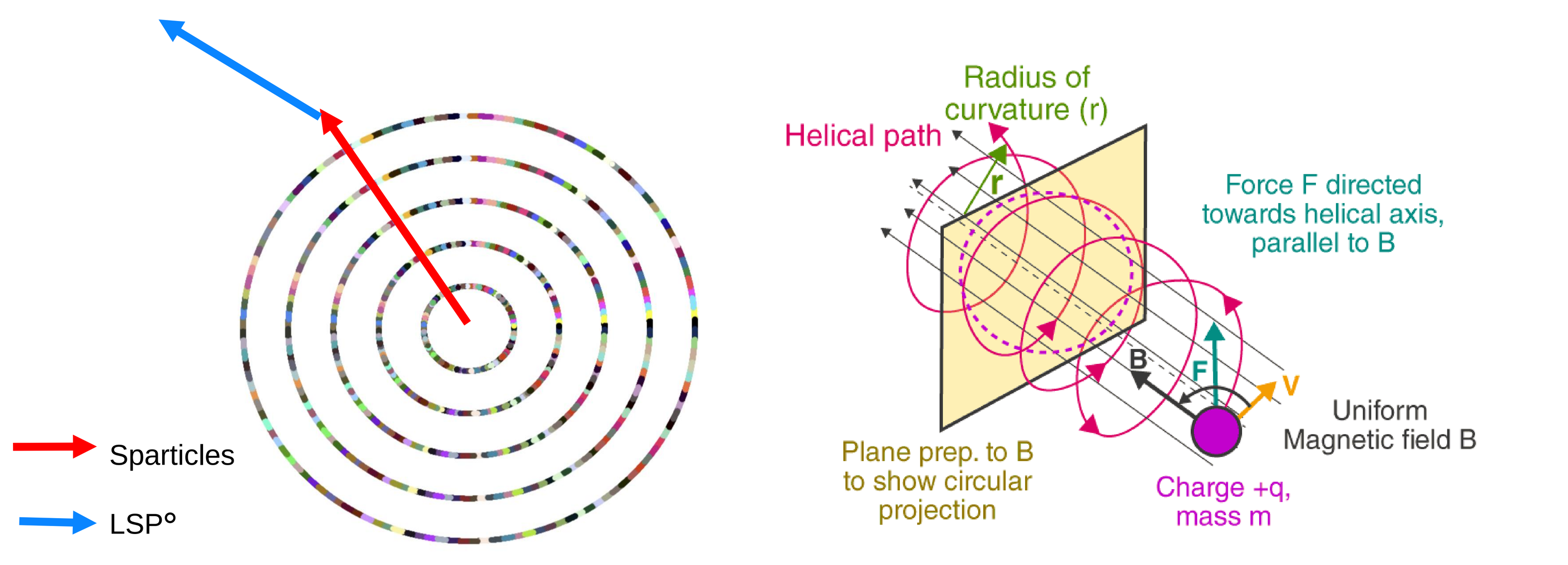

Fig: Schematic of transverse section (T.S.) of concentric detector circles ($5 \le r \le 25\ cm$) with colored wedge hits representing production of LSP° (left) and the helical trajectory of a charged particle in magnetic field (right, *source: Byjus*)

Physics Motivation:
The effect of dark matter at galactic and cosmological scales have been observed but they remain unincorporated in the Standard Model (SM) of physics.

Supersymmetric partners ($\chi^{\pm}$) of electroweak interaction mediators (W±) are theorized to decay into low energy pions and stable, neutral lightest supersymmetric particle (LSP°) which is a particulate dark-matter (DM) candidate.

$$\chi^{\pm} \longrightarrow \pi^{\pm} + LSP°$$

The challenge is to identify χ-mediators within short interval of about $4\mu s$ at HL-LHC.

Wedge Generation and Spacepoint Divider Algorithm

A. Wedge Generator Algorithm:
- We look for particles with relativistic momentum $p \ge 10\ GeV/c$ and at magnetic field $B = 2\ Tesla$ that theory suggests as signatures of the Sparticles
- We divide the T.S. of concentric cylindrical detectors into 128 wedges with circular boundaries determined by relativistic equation for magnetic Lorentz force on a charged particle:

$$r = \frac{p}{0.3Bn}$$

- Each wedge is constructed such that it covers the entirety of the T.S. by geometry and containment of any charged particle, with p and B as described, in a single wedge is ensured

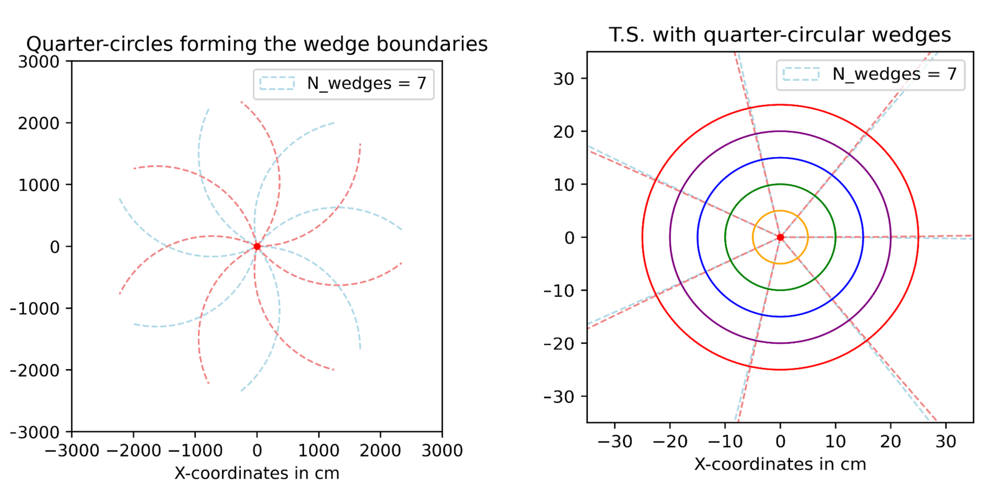

Fig: Transverse Geometry of wedge boundaries as quarter-circular arcs (left) and zoomed-in wedge boundaries with 5 concentric detector cylinders (right) for $N\_wedges = 7$

B. Spacepoint Divider Algorithm:
- In each wedge object, the spacepoints in the detector data is divided for all 128 wedges
- Spacepoints of each wedge is later analyzed for cover-making in the longitudinal section (L.S.) of the detectors

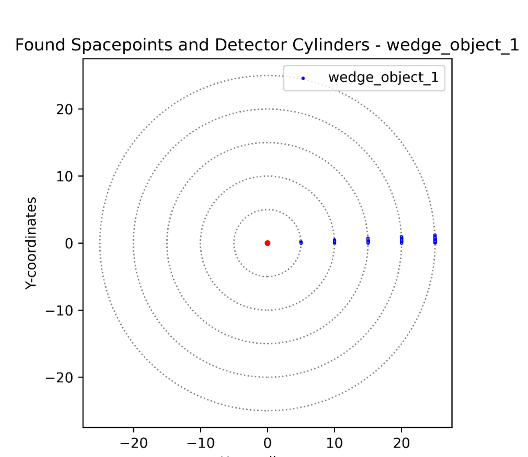

Fig: Transverse geometry of spacepoints divided into the first wedge object. These spacepoints will serve as data for the cover-making algorithm wedge-by-wedge.

Cover Generator Algorithm

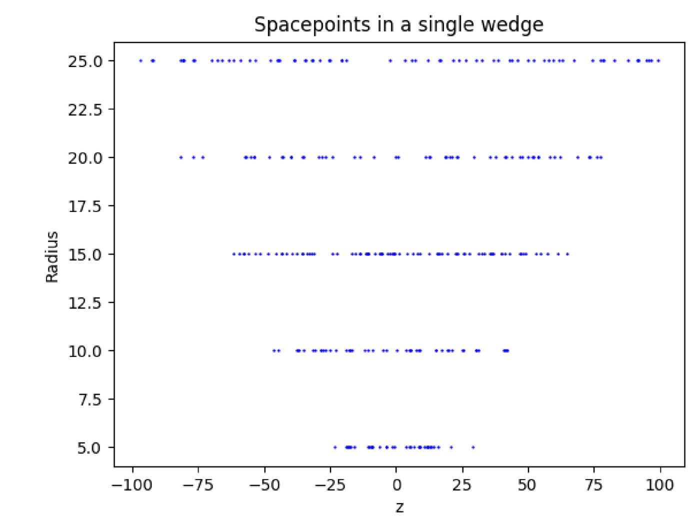

Fig: Longitudinal section (L.S.) of the concentric detector cylinders with distribution of spacepoints about the collision axis (z-axis)

The spacepoints are clustered into patches of 5 superpoints, each containing 16 spacepoints with a goal of containing a helix entirely in only one patch. The set of all the patches is called "cover."

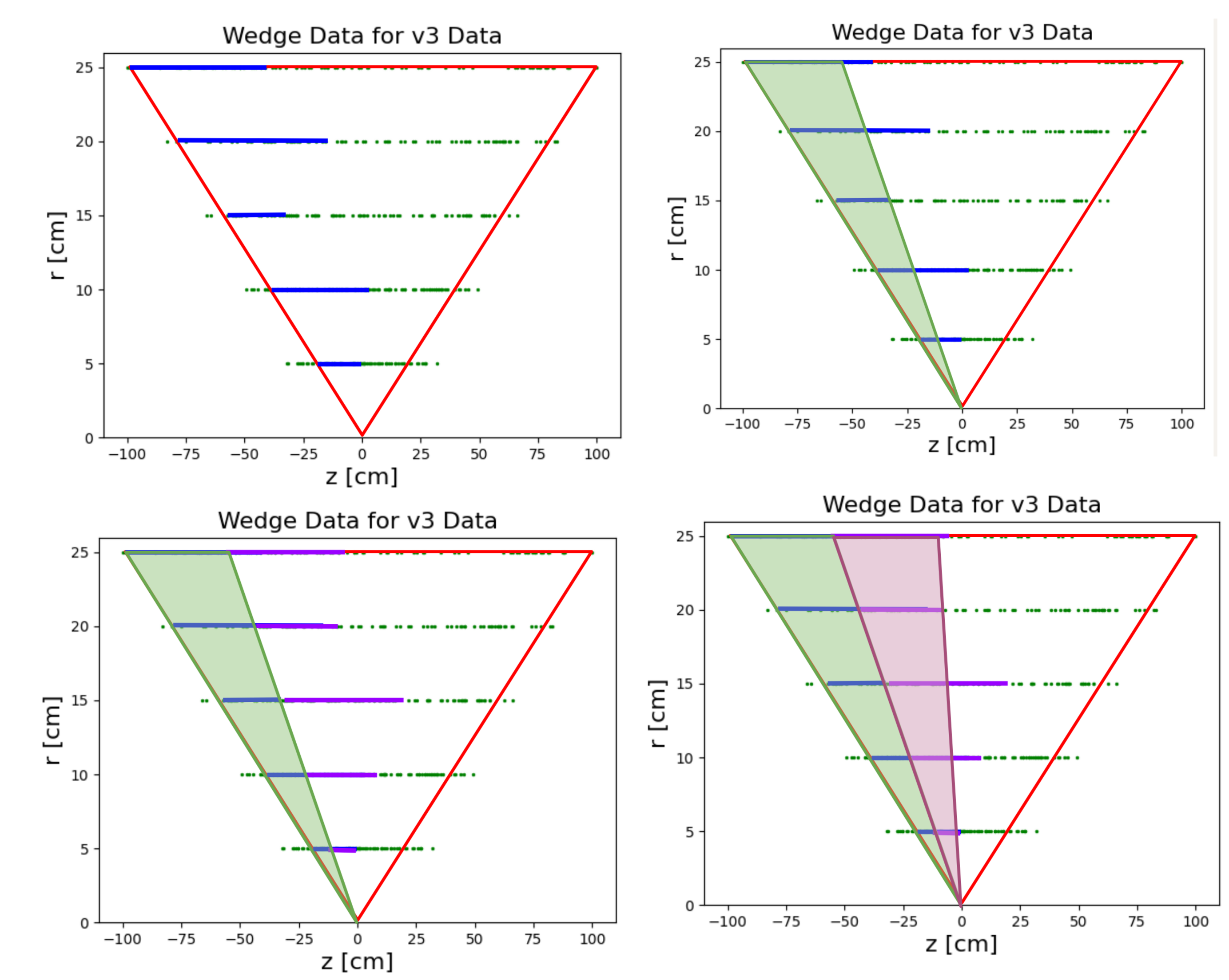

Fig: The process of spacepoint clustering to make patches as per the circuitry requirements to rapidly detect helices of charged particles

Helix Detecting Algorithm

- Helical trajectories are characterized by the azimuthal $\phi(r)$ and longitudinal $z(r)$ coordinates
- Each spacepoint is considered a node of a graph and is linked to its adjacent nodes
- Second derivatives of an assigned graph is calculated to develop a graph computing algorithm (see Ref 1. and 2. for details)
- It then identifies the geometry of the helix created by a signal particle
- This algorithm is to be embedded in a graph computing architecture using field programmable gate arrays (FPGAs)

Key Achievement:
We have shown that our method is amenable to a "first-level" track trigger with a latency less than $1\mu s$, which is compatible with $4\mu s$ upper limit set by the HL-LHC experiments.

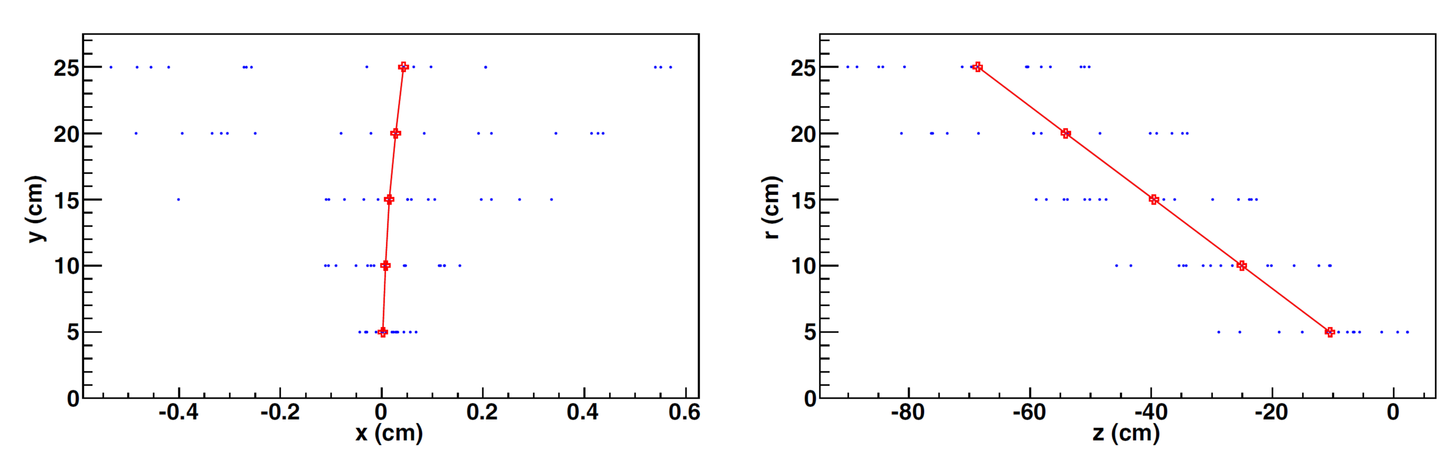

Fig: Example of an event showing the high-momentum signal track (red curve) found amongst pileup spacepoints (blue dots) in the T.S. (left) and L.S. (right). The red points denote hits created by a signal particle (Figure reproduced from Ref. 2)

Robustness Testing

We are applying several different methods to test the robustness of our algorithm with different versions of simulated detector data. The goal is to look for inconsistencies in results, identify the cause (mathematical, physical, or algorithmic), and incorporate solutions to make the algorithm more robust.

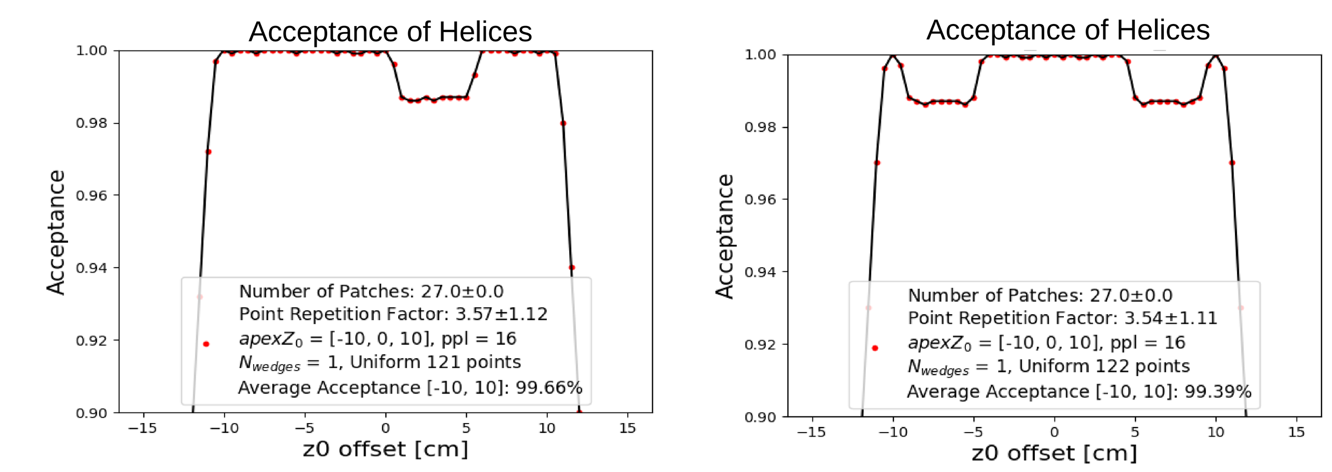

Fig: Plots of averaged acceptance values (containment of a helix completely in one patch) over the particle luminosity range from $-15\ to\ +15$ cm

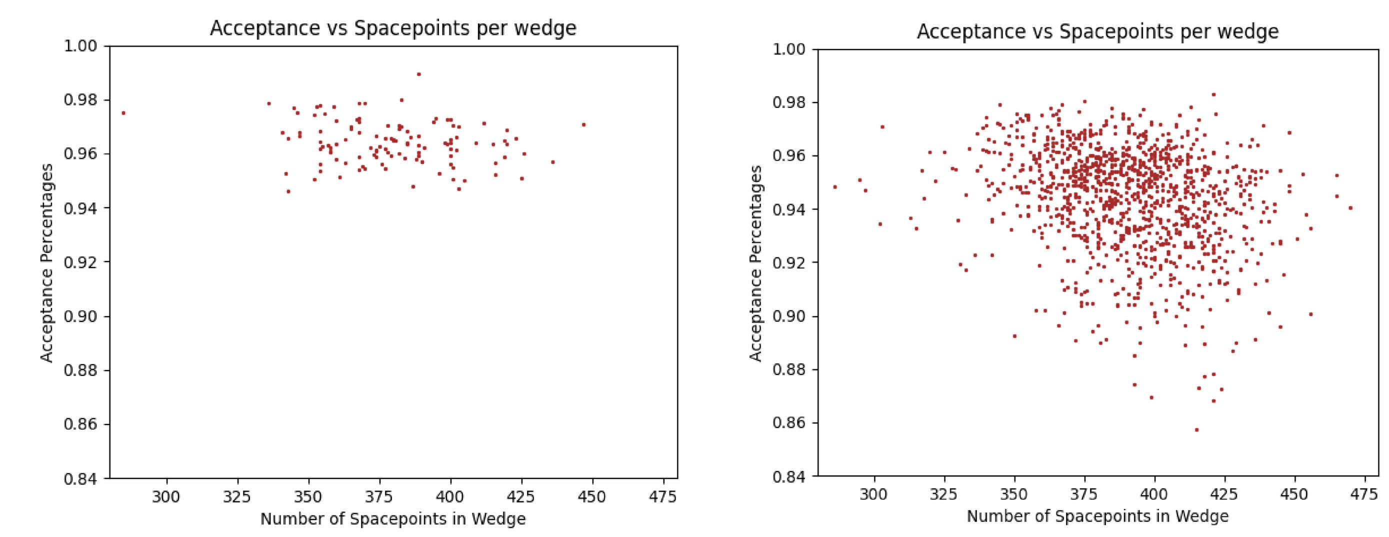

Fig: Plots of averaged acceptance percentages vs. total number of spacepoints in a good wedge data (left) and bad wedge data (right)

Conclusion

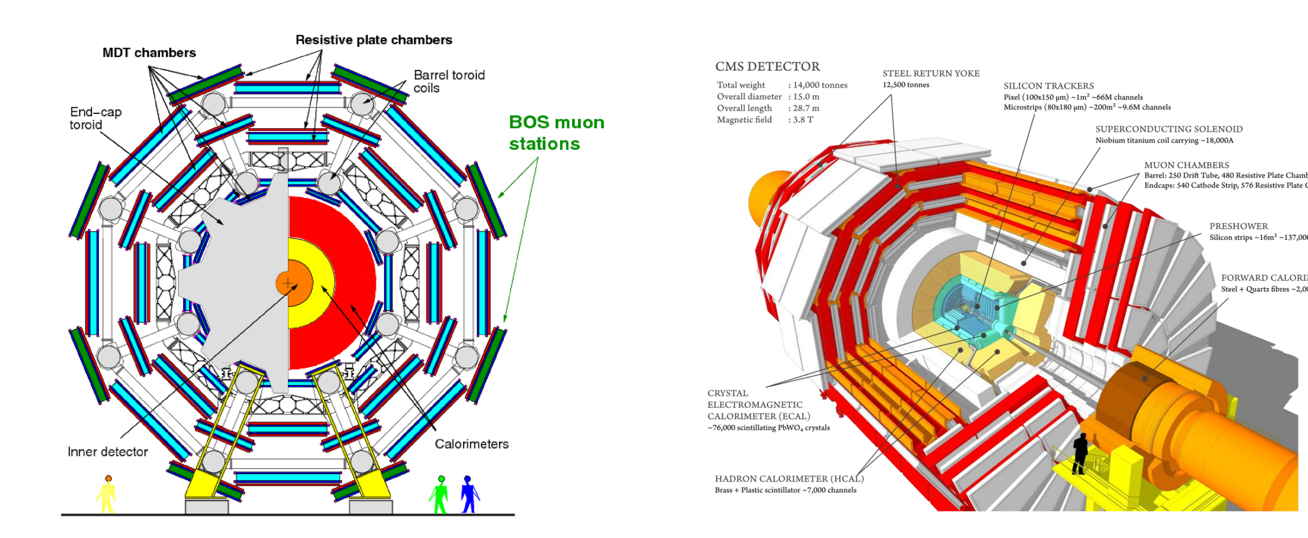

Fig: Representative image of the T.S. (left) and L.S. (right) of concentric detector cylinders at CMS experiment HL-LHC (*Source: CERN*)

- **Rapid reconstruction of charged-particle trajectories at the High Luminosity-Large Hadron Collider (HL-LHC) can be achieved by silicon-based track triggering.**
- **LHC may serve as a DM factory through copious production of heavier, charged partners of the theorized DM particles.**
- **Detection of such DM particles or other heavier new particles can be assisted through augmentation or replacement of traditional computing algorithms with our novel algorithm.**
- **Further testing and improvements are in process before embedding the algorithm into chips using commercial FPGAs.**

Acknowledgements & References

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